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MILITARY SPECIFIC ATJON

TEST SET, AIR DEFENSE ARTILLERY, AN/MWM-3; 8438880

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Air Defense Test Set, AN/MWM-3 to be used to troubleshoot and verify performance of components of the M163 Self-propelled and the M167 Towed, 20 Millimeter, Air Defense Artillery Guns known as the Vulcan Air Defense System, (VADS) . The test set includes a test set panel, a jack field panel, and associated cables located for protection within a transit case to be used to test each weapon system control panel, distribution box, sight current generator, and servo amplifiers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, term a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-W-13855	Weapons: Small Arms and Aircraft Armament Sub-systems? General Specification for
MIL-F-13926	Fire Control Materiel: General Specication Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of

FSC: 4931

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-I-45607	Inspection Equipment Acquisition Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

PUBLICATIONS

MIL-HDBK-472	Maintainability Prediction Handbook

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

8438880	Test Set, ADA, AN/MWM-3
8438956	Extender Board Assembly
8438963	Auxiliary Position Indicator Assembly

PACKAGING DATA SHEET

8438880	Packaging Data Sheet for Test Set, Air Defense Artillery, AN\MWM-3

(Copies of specification, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Test Set, Air Defense Artillery, AN/MWM-3, hereinafter referred to as test set, shall be manufactured in accordance with Drawing F8438880 and drawings pertaining thereto.

3.2 General specification. Where applicable, the test set shall meet the following requirements of MIL-W-13855, MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-W-13855

a. Marking
b. Workmanship

3.2.2 MIL-F-13926

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Soldering electrical connections
d. Part identification

3.2.3 MIL-C-46357

a. Etched circuit assembles
b. Wiring
c. Threaded fasteners

3.3 Environmental.

3.3.1 Vibration (Operating). The test set shall be capable of withstanding vibration with peak acceleration levels of 2.5g's over a frequency range of 7 Hz to 200 Hz in each of three mutually perpendicular axes without being damaged or adversely affected. The duration of vibration along each axis shall be 84 minutes. Subsequent to the vibration test, the test set shall not have been damaged or adversely affected and shall meet all the requirements of 3.5.

3.3.2 Altitude (operating). The test set shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.581 inches Hg).

3.3.2.1 Altitude (nonoperating). The test set shall meet the requirements of 3.4. and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 50,000 feet above sea level (3.444 inches Hg).

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3.3.3 Temperature (operating). The test set shall meet the requirements of 3.4 and 3.5 when thermally stabilized at any temperature between -400 Centigrade (-40°F) and +55°C (+130°F). Test sets shall intermittently operate at +71°C (+160°F) for on and off periods of 20 minutes duration.

3.3.3.1 Temperature (nonoperating, storage). The test set shall not suffer permanent damage when subjected to the temperature extremes of -62°C (-80°F) to +85°C (+185°F) and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.4 Humidity (nonoperating). The test set shall meet the requirements of 3.4 and 3.5 after exposure (case open) to a humidity of ten 24 hour cycles varying from 75% humidity at +55°C (+130°F) to 95% humidity at +32°C (+90°F), including conditions wherein condensation takes place in and on the equipment.

3.3.5 Immersion. The test set shall not leak when immersed in water a minimum of one inch. This test shall be performed with case closed and shall meet the requirements of 3.4 and 3.5 upon removal.

3.3.6 Fungus. The test set shall not be damaged nor shall its performance be impaired when subjected to the micro-organisms specified in Table I.

TABLE 1. MICRO-ORGANISMS

FUNGI	ATCC 1	NLABS[2]
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funciculosum	11797	474
Chaetomium globosum	6205	459

NOTE: 1. American Type Culture Collection.
2. Pioneering Research Division, U.S. Army Natick Laboratories.
3. Test duration length shall be 28 days.

3.3.7 Electromagnetic Interference Control. The test set shall comply with the requirements of MIL-STD-461 for Class IC: test RE02 (150 KHz to lGHz); RS03; (10 KHz to 400 MHz); and CS06 only.

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3.3.8 Reliability. The test set shall have a minimum reliability of 100 hours Mean Time Between Failures (MTBF).

3.4 Electrical Continuity. Each conductor shall show electrical continuity as specified on the schematic diagram.

3.5 Performance. The test set shall meet the requirements of this section after the input power has been applied to the test set for a period of not less than five minutes.

3.5.1 Controls. The test set switches and indicators shall meet the functions and requirements in Table II.

TABLE II. SWITCHES AND INDICATORS

	SWITCH OR INDICATOR	FUNCTION
a.	DRUM EMPTY indicator	When lighted, shall indicate weapon system last ROUND switch is closed, signaling that the drum is empty of ammunition.
b.	DIGITAL voltmeter	Shall monitor voltages from the weapon system as required, and present a digital readout.
c.	SELF CHECK push-button switch	When depressed, shall enable GO logic circuits to be self-checked.
d.	GO indicator	When lighted, shall indicate proper functioning of pre-selected weapon system relays in the distribution box .
e.	NO-GO indicator	When lighted, shall indicate non-acceptable functioning of pre-selected weapon system relays in the distribution box.
f.	RELAY rotary switch	Shall provide a selection of distribution box relay signals to be monitored by the GO and NO-GO indicators, or the digital voltmeter.

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TABLE II. SWITCHES AND INDICATORS (CONT'D)

SWITCH OR INDICATOR	FUNCTION
g. RESET push-button switch	When depressed, shall provide a +5 vdc reset signal to the logic circuits of the test set.
h. PROBLEM rotary switch	Shall provide selection of problem voltages applied by the test set to the weapon system sight current generator. Shall also provide routing for voltage from the sight current generator to RADAR antenna control.
i. SCG rotary switch	Shall provide selection of sight current generator voltages to be monitored by the digital volt-meter.
j. DVM-CLOCK rotary switch	Shall provide selection of SCG, SPIN TEST, or VOLTAGE positions as inputs to the digital voltmeter.
k. SPIN TEST/STOP toggle switch	Shall provide a manual means to start and stop the DVM clock during the weapon system spin test.
l. -VOLTAGES rotary switch	Shall provide selection of weapon system voltages as inputs to the digital voltmeter.
m. PWR ON\OFF switch	When set in ON position, shall provide circuit overload protection and shall enable weapon system battery power to be applied to the test set.
n.. TEST SET indicator	When lighted, shall indicate that weapon system battery power is applied to the test set.

3.5.2 The test set shall indicate the proper values within the tolerances specified in paragraphs 3.5.2.1 through 3.5.2.45.

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3.5.2.1 When 22 to 30 Vdc is applied to terminals J1-Y (+) and J1-e(-) and the front panel switches are set in the following manner; VOLTAGES to SYSTEM, DVM-CLOCK to VOLTAGES, POWER to ON, the DVM shall indicate the input voltage with an accuracy of ± 5%.

3.5.2.2 When the RELAYS switch is set to K6 and the POWER switch is set to ON, the indicator lamps, on the front panel of the test set, shall light for the duration of the time that the lamp heads are depressed.

3.5.2.3 When the DVM - CLOCK switch is set to V EXT and a jumper wire is connected between TP-2 and TP-GND and the power switch is set to ON, the DVM shall. indicate 0.000 ± 0.008 Vdc.

3.5.2.4 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to +20 and the power switch is set to ON, the DVM shall indicate +20 + 0.06 Vdc. An external voltmeter connected between J3-K(+) and J3-X(-) shall indicate +20 ± 0.02 Vdc.

3.5.2.5 When the DVM - CLOCK switch is set to S.T. RAMP and the POWER switch is set to ON and the RESET switch is depressed and released, the DVM shall indicate 0.000 + 0.002. This reading shall be adjustable by operation of the ZERO ADJUST potentiometer.

3.5.2.6 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to +5 and the POWER switch is set to ON, the DVM shall indicate +5.0 ± 0.5 Vdc.

3.5.2.7 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to +15 and the POWER switch is set to ON, the DVM shall indicate +15.0 ± 1% Vdc.

3.5.2.8 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to -15 and the POWER switch is set to ON, the DVM shall indicate -15.0 ± 1% Vdc.

3.5.2.9 When the DVM - CLOCK switch is set to V EXT and +40.0 + 0.005 Vdc is applied to TP-2(+) and TP-GND (-) and the POWER switch is set to On the DVM shall. indicate +39.8 to +40.2 Vdc.

3.5.2.10 When the front panel switches are set as in 3.5.2.9 and +0.050 + 0.0005 Vdc are applied at TP-2(+) and TP-GND (-), the DVM shall indicate +0.046 to +0.054 Vdc.

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3.5.2.11 When the front panel switches are set as in 3.5.2.9 and -40.0 + 0.005 Vdc are applied at TP-2(+) and TP-GND(-), the DVM shall indicate -39.8 to -40.2 Vdc.

3.5.2.12 When the front panel switches are set as in 3.5.2.9 and -0.050 ± 0.005 Vdc are applied at TP-2(+) and TP-GND(-), the DVM shall indicate -0.046 to -0.054 Vdc.

3.5.2.13 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 8 and the POWER switch is set to ON, the DVM shall indicate +20.0 ± 0.3% Vdc.

3.5.2.14 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 7 and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - J(+) and J3 - X(-) shall indicate +10.62 ± 1% Vdc.

3.5.2.15 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 6 and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3-H (+) and J3-X (-) shall indicate -10.62 ± 1% Vdc.

3.5.2.16 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 5 and the PROBLEM switch is set to 1, 2, 4, 5 and 7 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - AA (+) and J3-X (-) shall indicate -5.31 ± 1% Vdc for each of the designated positions of the PROBLEM switch indicated above.

3.5.2.17 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 5 and the PROBLEM switch is set to 3 and 8 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - AA(+) and J3-X (-) shall indicate - 9.46 ± 1% Vdc for each of the designated positions of the PROBLEM switch indicated above.

3.5.2.18 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 5 and the PROBLEM switch is set to 6 and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - AA(+) and J3 X (-) shall indicate +5.31 ± 1% Vdc.

3.5.2.19 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 4 and the PROBLEM switch is set to 1 and 3 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - X(+) and J3 - X (-) shall indicate +2.0 ± 0.3% Vdc for each of the designated positions of the PROBLEM switch indicated above.

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3.5.2.20 When the DVM - CLOCK switch is set to SCG 1 and the SCC switch is set to 4 and the PROBLEM switch is set to 2 and 6 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - x (+) and J3 - X (-) shall indicate +4.0 + 0.3% Vdc for each of the designated positions of the PROBLEM switch indicated above.

3.5.2.21 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 4 and the PROBLEM switch is set to 4 and 5 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3 - x (+) and J3 - X (-) shall indicate +8.0 + 0.3% Vdc for each of the designated positions of the PROBLEM switch indicated above.

3.5.2.22 When the DVM - CLOCK switch is set to SCG 1 and the SCG switch is set to 4 and the PROBLEM switch is set to 7 and 8 consecutively and the POWER switch is set to ON, the DVM and an external voltmeter connected between J3-x(+) and J3-X(-) shall indicate + 10.0 + 0.3% Vdc for each of the designated positions of the PROBLEM switch indicated above.

3.5.2.23 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to FV1 and + 40.0 ± 0.005 Vdc are applied between J1-m (+) and J1 - e (-) and the POWER switch is set to ON, the DVM shall indicate +0.40 + 0.04 Vdc.

3.5.2.24 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to 1 Ø and 115 + 5% VAC, 400 + 5% Hz is applied between J1-c (+) and J1-d (-) with DC Power Supply ground J1-e, connected to TP Grid, and the power supply switch is set to ON, the DVM shall indicate + 10.0 to + 12.5 Vdc.

3.5.2.25 When the DVM – CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to AC - SCG and 115 + 5% VAC, 400 + 5% Hz is applied between J3-A (+) and J3 - B (-) with the DC Power Supply ground J1-e, connected to TP Ground and the power switch set to ON, the DVM shall indicate + 10.0 to + 12.5 Vdc.

3.5.2.26 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to ØA and 115 ± 5% VAC, 400 + 5% Hz is applied between J1-V(+) and J1-U(-) with the DC Power Supply Ground J1-e, connected to TP Ground and the power switch set to ON, the DVM shall indicate + 10.0 to + 12.5 Vdc.

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3.5.2.27 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to ØB and 115 ± 5% VAC, 400 + 5% Hz is applied between J1-V(+) and J1-U(-) with DC Power Supply Ground J1-e, connected to TP Ground and the power switch set to ON, the DVM shall indicate + 10.0 to + 12.5 Vdc.

3.5.2.28 When the DVM - CLOCK switch is set to VOLTAGES and the VOLTAGES switch is set to PC and 115 ± 5% VAC, 400 + 5% Hz is applied between J1-W(+) and J1-U(-) with DC Power Supply Ground J1-e, connected to TP Ground and the power switch set to ON, the DVM shall indicate + 10 to + 12.5 Vdc.

3.5.2.29 when the RELAYS switch is set to K6 and the POWER switch is set to ON, the GO indicator lamp shall light when the SELF CHECK switch is depressed.

3.5.2.30 When the RELAYS switch is set to K10 and the POWER switch is set to ON, the GO indicator lamp shall light when the SELF CHECK switch is depressed.

3.5.2.31 When the RELAYS switch is set to LO RATE and the POWER switch is set to ON, the NO-GO indicator lamp shall light and remain lit. When the RESET switch is depressed and released, the NO-GO indicator lamp shall extinguish. When the SELF-CHECK switch is depressed and released, the GO indicator lamp shall light and remain lit. The GO indicator lamp shall extinguish when the RESET switch is again depressed and released.

3.5.2.32 When the RELAYS switch is set to HI RATE and the POWER switch is set to ON, the NO-GO indicator lamp shall light and remain lit. When the RESET switch is depressed and released, the NO-GO indicator lamp shall extinguish. When the SELF-CHECK switch is pressed and released, the GO indicator lamp shall light and remain lit. The GO indicator lamp shall extinguish when the RESET switch is again depressed and released.

3.5.2.33 When the DVM Clock Switch is set to VOLTAGES and the VOLTAGES switch is set to FV-2 and the input power voltages to the test set is adjusted to + 27.5 + 0.5 Vdc and the power switch is set to ON, the DVM shall indicate 3.4 + 0.08 when the SELF-CHECK switch is depressed and held.

3.5.2.34 When the RELAYS switch is set to K5 and the POWER switch is set to ON (with the normal input power voltage energizing the test set), and a multimeter in the milliamp position connected between J1-H(+) and J1-e(-), the multimeter shall indicate less than 1.0 milliamp when the SELF-CHECK switch is depressed.

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3.5.2.35 With the same conditions existing as in 3.5.2.34, except that the multimeter is connected between J1-J(+) and J1-e(-), the multimeter shall indicate less than 1.0 milliamp when the SELF–CHECK switch is depressed.

3.5.2.36 With the same conditions existing as in 3.5.2.34, except that the multimeter is connected between J1-S(+) and J1-e(-), the multimeter shall indicate less than 1.0 milliamp when the SELF-CHECK switch is depressed.

3.5.2.37 With the same conditions existing as in 3.5.2.34, except that the multimeter is connected between J1-t(+) and J1-e(-), the multimeter shall indicate less than 1.0 milliamp when the SELF-CHECK switch is depressed.

3.5.2.38 With the same conditions existing as in 3.5.2.34, except that the multimeter is connected between J1-B(+) and J1-e(-), the multimeter shall indicate less than 1.0 milliamp when the SELF-CHECK switch is depressed.

3.5.2.39 When the RELAYS switch is set to LO RATE, and the DVM CLOCK switch is set to LO GUN DRIVE, and 27.5 +0.5 Vdc is applied between J1-t(+) and J1-e(-) through a NORMALLY OPEN test switch (SI), and 27.5 Vdc is applied between J1-j(+) and J1-e(-) through a NORMALLY CLOSED test switch (S2), and when the RESET switch is depressed and released, the DVM shall indicate 0.000 + 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. When test switch S1 is momentarily set to ON then to OFF, and test switch S2 is set to OFF immediately when the NO–GO indicator lights, the DVM shall indicate 3.0 to 4.5 Vdc.

3.5.2.40 When the RELAYS switch is set to LO RATE, and the DVM-CLOCK switch is set to CLR, and 27.5 Vdc is applied between J1-T(+) and J1-e(-) through a NORMALLY OPEN test switch (S1), and 27.5 + 0.5 Vdc is applied between J1-J(+) and J1-e(-) through a NORMALLY CLOSED test switch (S2) and when the RESET switch is depressed and released, the DVM shall indicate 0.000 + 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. When test switch S1 is momentarily set to ON then to OFF and test switch S2 is set to OFF’ immediately when the NO-GO indicator lights, the DVM shall indicate 3.0 to 4.5 Vdc.

3.5.2.41 When the RELAYS switch is set to LO RATE, and the DVM-CLOCK switch is set to F.V., and 27.5 + 0.5 Vdc is applied between J1-T(+) and J1-e(-) through a NORMALLY OPEN test switch (S1), and 27.5 Vdc is applied between J1-S(+) and J1-e(-) through a NORMALLY CLOSED test switch (S2), and when the RESET

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switch is depressed and released, the DVM shall indicate 0.000 + 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. When test switch S1 is momentarily set to ON then to OFF, and test switch S2 is set to OFF immediately when the NO-GO indicator lights, the DVM shall indicate 3.0 to 4.5 Vdc.

3.5.2.42 When the RELAYS switch is set to LO RATE, and the DVM-CLOCK switch is set to B.L., and 27.5 ± 0.5 Vdc is applied between J1-B(+) and J1-e(-) (in addition to J1-t) through a NORMALLY OPEN test switch (S1), and when the RESET switch is depressed and released, the DVM shall indicate 0.000 + 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. When test switch S1 is switched to ON for 2.0 + 0.3 seconds then OFF, the DVM shall indicate 1.0 to 3.0 Vdc.

3.5.2.43 When the RELAYS switch is set to K9 and the DVM - CLOCK switch is set to S.T. RAMP and an oscilloscope is connected between TP-1 (+) and TP-GND (-). The DVM shall indicate 0.000 ± 0.002 Vdc when the RESET switch is depressed and released. This reading shall be adjustable with the ZERO ADJUST potentiometer. The DVM shall indicate 3.5 ± 0.8 Vdc. and the oscilloscope shall indicate a pulse width of 3.5 + 0.8 seconds duration, when the SELF-CHECK switch is depressed and released.

3.5.2.44 When the DVM - CLOCK switch is set to SPIN TEST and the RESET switch is depressed and released, the DVM shall indicate 0.000 ± 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. When the SPIN TEST switch is set to TEST and the SELF CHECK switch is depressed and 8 + 0.25 seconds are allowed to elapse and then the SPIN TEST switch is set to STOP, the DVM shall indicate 8.0 ± 0.5 Vdc.

3.5.2.45 When 115 ± 5% VAC, 400 + 5% Hz is applied between J1c and J1-d (-) and the Auxillary position Indicator (8438963) is connected to Test Set connector J4 with cable W34 and the switch inside the APU is set to AZIMUTH and the PROBLEM switch on the test set is set to 4, the APU shall indicate 192° 20′ to 192° 40'. When the PROBLEM switch is set to 3, the APU shall indicate 167° 20′ to 167° 40′ . When the switch inside the APU is set to ELEVATION and the PROBLEM switch on the Test Set is set to 5, the APU shall indicate 167° 20′ to 167° 40′. When the PROBLEM switch is set to 6, the APU shall indicate 192° 20′ to 192° 40′.

3.6 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7, when tested in accordance with 4.7.

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3.7 Marking. The test set shall be clearly marked in accordance with MIL-W-13855. The identification plate shall be fastened securely to the test set as specified on the drawing. Unless otherwise specified (see 6.1), each test set shall be identified by a serial number assigned by the procuring activity.

3.8 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Classification of inspections. Unless otherwise specified herein or in the contract, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provision of 4.3.2. Unless otherwise specified, the sample for first article approval tests shall consist of three test sets plus three each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The responsibility (Government or Contractor) for first article testing shall be as specified in the contract.

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4.3.2 Inspections to be performed.

4.3.2.1 Test Sets. First article test sets shall be tested as specified herein for all the requirements of this specification. First article test shall be performed on the initial production items. Three samples shall be selected by the Government. First article test will not be performed again provided the manufacturing processes or techniques used to produce the samples have not been changed or revised that would, in the opinion of the contracting officer, have an adverse effect on the remaining quantities produce.

4.3.2.2 SQAP items. The items covered by SQAPS shall be tested as specified therein, for all characteristics delineated in the SQAP.

4.3.3 Rejection. If any test set or SQAP item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component or subassembly in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.4.2 Examination. The examinations listed herein for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the Supplementary Quality Assurance Provisions (SQAP) listed in the Technical Data Package (TDP). In the absence of SQAP’S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4.2.2 Test sets. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, Inspection Level 11, and the sampling plans specified in Tables III herein. Examinations and

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tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated defects in Tables III, IV, V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE 111. CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
101	Fabrication	3.1	Applicable Dwg: and Visual
102	General Specification	3.2	MIL-F-13926 and MIL-C-46357 Visual
103	Vibration	3.3.1	4.6.1.1
104	Continuity	3.4	Tactile & 4.6.2.1
105	Setting & Function of Test SW.	3.5.1	Visual & Tactil
106	Relay SW set to K6 Position	3.5.2.2	4.6.3.3
107	Clock SW set to V. Ext.	3.5.2.3	4.6.3.4
108	Clock SW set to Voltage	3.5.2.4	4.6.3.5
109	Clock SW set to S.T. Ramp	3.5.2.5	4.6.3.6
110	Clock SW set to +5.0 V	3.5.2.6	4.6.3.7
111	Clock SW set to +15 V	3.5.2.7	4.6.3.8
112	Clock SW set to -15 V	3.5.2.8	4.6.3.9
113	Clock SW set to V Ext. (w/+40 Vdc applied to TP-2(+) and TP-Ground (-))	3.5.2.9	4.6.3.10
114	Clock SW set to V. Ext. (w/4.050 Vdc Applied to TP-2(+) and TP Ground (-)	3.5.2.10	4.6.3.11
115	Clock SW set to V. Ext. (w/-40 Vdc Applied to TP-2(+) and TP Ground (-))	3.5.2.11	4.6.3.12
116	Clock SW set to V. Ext. (w/-050 Vdc Applied to TP-2(+) and TP Ground (-)	3.5.2.12	4.6.3.13

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TABLE III. CLASSIFICATION OF DEFECTS (CONT'D)

CLASS CRITICAL: MAJOR:	CHARACTERISTIC NONE AQL 0.65% Defective	REQUIREMENT	TEST PROCEDURE
117	Clock SW set to SCG 1 and SCG SW set to 8	3.5.2.13	4.6.3.14
118	Clock SW set to SCG 1 and SCG SW set to 7	3.5.2.14	4.6.3.15
119	Clock SW set to SCG 1 and SCG SW set to 6	3.5.2.15	4.6.3.16
120	Clock SW set to SCG 1 and SCG SW set to 5 Problem SW set to 1-2-4-5-7 (in that order)	3.5.2.16	4.6.3.17
121	Clock SW set to SCG 1, SCG SW set to position 1 and 5, Problem SW set to 3 and 8 (in that order)	3.5.2.17	4.6.3.18
121	Clock SW set to SCG 1, SCG SW set to position 5, Problem SW set to 6	3.5.2.18	4.6.3.19
123	Clock SW set to SCG 1, SCG SW set to position 4, Problem SW set to 1 and 3 (in that order)	3.5.2.19	4.6.3.20
124	Clock SW set to SCG 1, SCG SW set to position 4, Problem SW set to 2 and 6 (in that order)	3.5.2.20	4.6.3.21
125	Clock SW set to SCG 1, SCG SW set to position 4, Problem SW set to 4 and 5 (in that order)	3.5.2.21	4.6.3.22
126	Clock SW set to SCG 1, SCG SW set to position 4, Problem SW set to 7 and 8 (in that order)	3.5.2.22	4.6.3.23
127	Clock SW set to Voltage-Voltage SW set to FV1 w/+40.0 Vdc applied	3.5.2.23	4.6.3.24
128	Clock SW set to Voltage-Voltage SW set to 1Ø and 115 Vdc applied	3.5.2.24	4.6.3.25

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TABLE 111. CLASSIFICATION OF DEFECTS (CONT'D)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
129	Clock SW set to Voltage-Voltage SW set to AC and 115 V. applied	3.5.2.25	4.6.3.26
130	Clock SW set to Voltage-Voltage SW set to ØA and 115 V. applied	3.5.2.26	4.6.3.27
131	Clock SW set to Voltage-Voltage SW set to ØB and 115 V. applied	3.5.2.27	4.6.3.28
132	Clock SW set to Voltage-Voltage SW set to ØC and 115 V. applied	3.5.2.28	4.6.3.29
133	Relay SW set to K6, Power SW on	3.5.2.29	4.6.2.30
134	Relay SW set to K10, Power SW on	3.5.2.30	4.6.2.31
135	Relay SW set to Lo-Rate, Power SW on	3.5.2.31	4.6.2.32
136	Relay SW set to Hi-Rate, Power SW on	3.5.2.32	4.6.3.33
137	Clock SW set to Voltage-Voltage SW set to FV-2	3.5.2.33	4.6.3.34
138	Relay SW set to K-5, Power SW on and Multi-meter connected between J1-H(+) and J1-e(-)	3.5.2.34	4.6.3.35
139	Relay SW set to K-5, Power SW on and Multi-meter connected between J1-J(+) and J1-e(-)	3.5.2.35	4.6.3.36
140	Relay SW set to K5, Power SW on and Multi-meter connected between J1-S(+) and J1-e(-)	3.5.2.36	4.6.3.37
141	Relay SW set to K5, Power SW on and multi-meter connected between J1-t(+) and J1-e(-)	3.5.2.37	4.6.3.38

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TABLE III. CLASSIFICATION OF DEFECTS (CONT'D)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
142	Relay SW set to K5, Power SW on and multi-meter connected between J1 B(+) and J1-e(-)	3.5.2.38	4.6.3.39
143	Relay SW set to Lo-Rate, Clock SW set to Lo Gun Drive and 27.5 Vdc applied to J1-t(+) and J1-e (-)	3.5.2.39	4.6.3.40
144	Relay SW set to Lo-Rate, Clock SW set to C/R and 27.5 Vdc applied to J1-t(-t) and J1-e(-)	3.5.2.40	4.6.3.41
145	Relay SW set to Lo-Rate, Clock SW set to F.V. and 27.5 Vdc applied to J1-t(+) and J1-e(-)	3.5.2.41	4.6.3.42
146	Relay SW set to Lo-Rate, Clock SW set to B.L. and 27.5 Vdc applied to J1-t(+) and J1-e(-)	3.5.2.42	4.6.3.43
147	Markings	3.7	4.8
148	Workmanship	3.8	4.9
MINOR:	NONE		

NOTE : The tests in Table III shall be conducted at a temperature of 730 + 180F.

4.4.2.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Environmental test

4.5.1 Environmental. Three test sets shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table III.

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TABLE IV. CLASSIFICATION OF DEFECTS

CHARACTERISTIC	REQUIREMENT	TEST METHOD
201 Operating Temperatures (-40° and +130°F)	3.3.5	4.6.1.5 & 4.6.1.5.1
202 Storage Temperatures (-80° and +185°F)	3.3.5.1	4.6.1.5 & 4.6.1.5.1

4.5.1.1 Process verification (follow-up) . Three test sets shall be randomly selected as samples from production lots approximately at 25-50-75% completion interval of each production order or contract. Each sample shall meet the requirements and test in Table V.

TABLE V. CLASSIFICATION OF DEFECTS

CHARACTERISTIC	REQUIREMENT	TEST METHOD
301 Altitude (operating)	3.3.2	4.6.1.2
302 Altitude (nonoperating)	3.3.2.1	4.6.1.2
303 Humidity	3.3.4	4.6.1.4
304 Rain	3.3.5	4.6.1.5
305 Fungus	3.3.6	4.6.1.6 Note (1)
306 Electromagnetic Interference	3.3.7	4.6.1.7 Note (1)
307 Reliability	3.3.8	4.6.1.8 Note (1)
308 Interchangeability	3.6	4.7

*(Note (1) The requirements for Fungus, Humidity Electromagnetic Interference and Reliability need not be performed provided the manufacturing processes and-techniques have not been changed subsequent to First Article approval. Acceptance for noted characteristics shall be by certification.

4.5.1.2 Failure of sample. Should any one item fail to meet requirements specified in Table's III, IV and V, Acceptance of the Product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.4.2.3) .

4.5.2 Inspection equipment. Requirements for inspection equipment to perform examination and test are specified herein and in specific contract clauses as applicable. See Section VI of this specification for general provisions applicable to inspection equipment designs.

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4.5.3 Government furnished inspection equipment. Where the contract provides, Government furnished test equipment supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.4 Contractor furnished inspection equipment. Contractors test equipment utilized to perform acceptance inspection shall be comparable to the limits outlined in Table VI and in paragraph 6.3.2. Test equipment designs provided by the contractor shall be submitted for approval as specified in 6.3.3.

TABLE VI. TEST EQUIPMENT

ITEM	DESCRIPTION
(a) Extender Board	Dwg #8438956 or equivalent
(b) Oscilloscope	Dual Trace Textronix Model 454 or equivalent
(c) Digital Multimeter	Cimron Model E9500B-14 or equivalent
(d) Multimeter	Western Reserve Model 300M-A or equivalent
(e) Auxiliary Position Indicator Assembly	P/N 8438963 or equivalent
(f) D.C. Power Supply	Mid-Eastern Industries Inc. or equivalent

4.6 Test methods and procedures.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. The vibration test shall be performed in accordance with MIL-STD-810, Method 514.2, procedure X, Curve AX of Figure 514.2.7 and conditions specified in 3.3.1. Subsequent to the vibration test, the assembly shall be tested as specified in 4.6.2 and in 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Altitude (nonoperating and operating. This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 with the exception that the barometric pressures shall be as specified in paragraphs 3.3.2 and 3.3.2.1. The tests specified in 4.6.2 and 4.6.3 shall be performed as required to determine compliance with 3.3.2 and 3.3.2.1.

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4.6.1.3 High temperature (operating and nonoperating) This test shall be performed as specified in Procedure I, Method 501.1 of MIL-STD-810: The temperature of Step 2 shall be +185°F. The temperature of Step 2 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +160°F. In Step 5 the temperature shall be maintained for 4 hours and the assembly shall be tested in accordance with 4.6.2 and 4.6.3 during this time to determine conformance with paragraphs 3.3.3 and 3.3.3.1.

4.6.1.3.1 Low temperature (operating and nonoperating) . This test shall be performed as specified in Procedure I, Method 502.1 of MIL-STD-810. The temperature of Step 2 shall be -80°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the assembly shall be tested in accordance with 4.6.2 and 4.6.3 to determine compliance with paragraphs 3.3.3 and 3.3.3.1.

4.6.1.4 Humidity (nonoperating. This test shall be performed in accordance with Procedure I, Method 507.1 of MIL-STD-810, with exceptions of conditions as specified in Figure 1. The assembly shall be tested in accordance with 4.6.2 and 406.3 to determine compliance with paragraph 3.3.4.

4.6.1.5 Immersion. This test shall be performed in accordance with Procedure I, Method 51.2.1 of MIL-STD-810, with case closed. After completion of the test, the test set shall be tested in accordance with 4.6.2 and 4.6.3 to determine compliance with paragraph 3.3.5.

4.6.1.6 Fungus. One of the first five test sets produced shall be subjected to testing in accordance with Procedure I, Method 508,1 of MIL-STD-810. After completion of the test, the test set shall be tested in accordance with 4.6.2 and 4.6.3 to determine compliance with paragraph 3.3.6. After approval, subsequent acceptance for fungus requirements may be by certification.

4.6.1.7 Electromagnetic interference, This test shall be performed in accordance with Method RE02, Method RS03 and Method CS06 of MIL-STD-462 to determine compliance with paragraph 3.3.7.

4.6.1.8 Reliability demonstration. The test set shall be subjected to the total operating hours and acceptance criteria cited in Table VII to demonstrate the reliability requirement of paragraph 3.3.8. A minimum of four hours during each of the operating environments of paragraphs 3.3.2.1, 3.3.3.1, and 3.3.5

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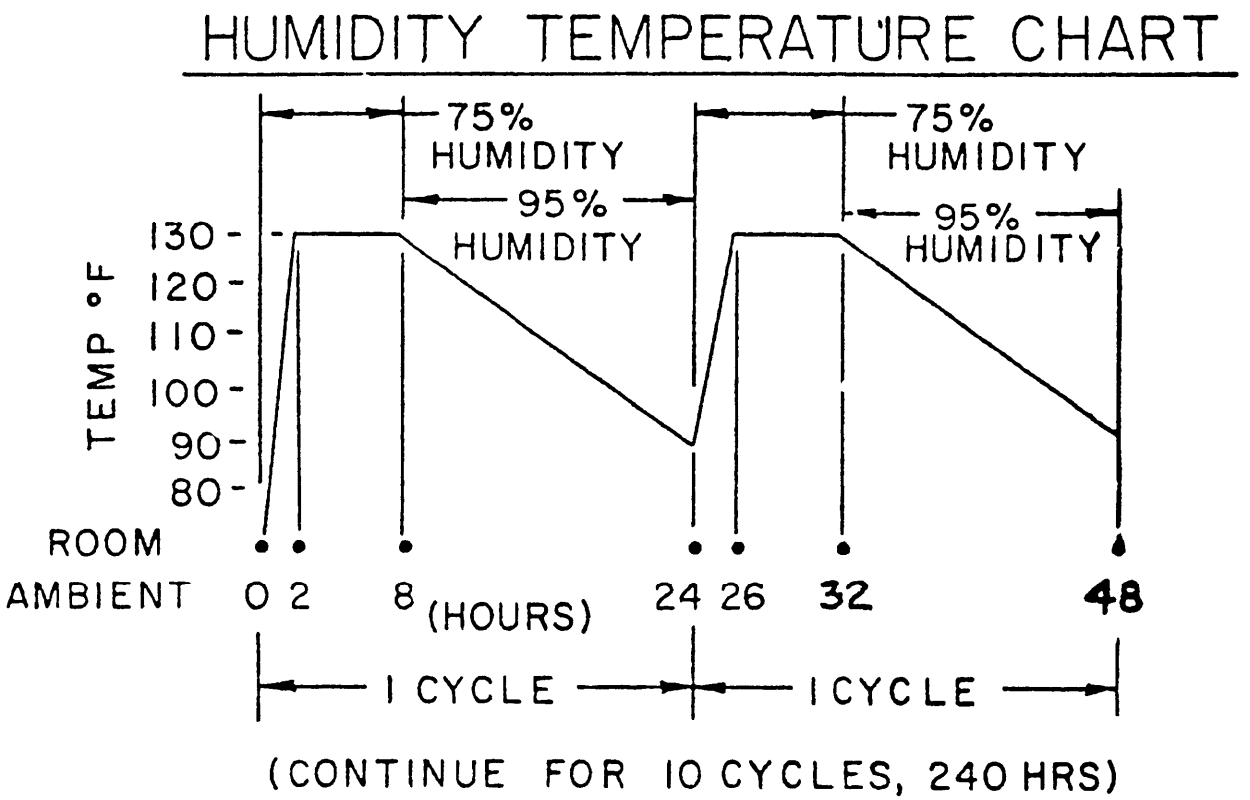


FIG. 1

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shall be accumulated on each test set subjected to the demonstration. A performance test, paragraph 4.6.2, and 4.6.3 shall be conducted each two hours of operating time. The mode of test set operation between performance tests shall be determined by successively preceding one step of the performance test for each interval. A test incident is defined as any failure of the test set to meet the performance requirements of this specification. A failure analysis shall be conducted on each test incident and all corrective actions documented. For production acceptance purposes a reliability failure is defined as any test incident which cannot be traced to inadequate design. In the event of a failure traced to inadequate design, the procuring activity (PCO) shall be notified prior to the initiation of corrective actions. Any reliability failure traced to materials, manufacturing processes, and/or workmanship must be corrected on the test set prior to resuming testing. Corrective action shall be taken on the material, process, and/or workmanship and all assembled or inprocess test sets. If any test incident occurs during the environment must be repeated. Reliability failure accountability will be determined by the procuring activity.

TABLE VII

TOTAL OPERATING HOURS	RELIABILITY FAILURES ACCEPT	REJECT
230	0	6
390	1	6
535	2	6
670	3	6
800	4	6
930	5	6
1055	6	

4.6.2 Electrical.

4.6.2.1 Continuity. Continuity shall be inspected by standard test equipment and visual examination for compliance with 3.4.

4.6.3 Performance.

4.6.3.1 Apply +22 to +30 Vdc in 4 volt increments to terminals J1-Y(+) and J1-e(-) and perform all tests specified in 4.6.3.2 through 4.6.3.46. After completion of testing at each voltage increment, the POWER switch shall be set to OFF and the POWER indicator shall extinguish.

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4 .6.3.2 Apply 22 to 30 Vdc to terminals J1-Y (+) and J1-e(-) and set the front panel switches in the following manner; VOLTAGES to SYSTEM, DVM - CLOCK to VOLTAGES, POWER to ON. The DVM shall indicate the input voltage within the limits specified in paragraph 3.5.2.1.

4.6.3.3 Set the RELAYS switch to K6 and the POWER switch to ON, the indicator lamps on the front panel of the test set shall illuminate for the duration of the time that the lamp heads are depressed as specified in paragraph 3.5.2.2.

4.6.3.4 Set the DVM - CLOCK switch to V EXT and connect a jumper wire between TP-2 and TP-GND and the power switch to ON. The DVM shall indicate a voltage within the limits specified in paragraph 3.5.2.3.

4.6.3,5 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to +20 and the power switch to ON. The DVM shall indicate +20 ± 0.06 Vdc. Connect an external voltmeter between J3-K(+) and J3-X(-). The voltmeter shall indicate the voltage specified in paragraph 3.5.2.4.

4.6.3.6 Set the DVM - CLOCK switch to S.T. RAMP and the POWER switch to ON. Depress and release the reset switch. The DVM shall indicate the value specified in paragraph 3.5.2.5. This reading shall be adjustable by operation of the ZERO ADJUST potentiometer .

4.6.3.7 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to +5 and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.6.

4.6.3.8 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to +15 and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.7.

4.6.3.9 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to -15 and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.8.

4.6.3.10 Set the DVM - CLOCK switch to V EXT and apply +40.0 + 0.005 Vdc to TP-2 (+) and TP-GND(-) and with the POWER switch set to ON the DVM shall indicate a reading within the limits specified in paragraph 3.5.2.9.

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4.6.3.11 Set the front panel switches as in 4.6.3.10 and apply +0.050 ± 0.0005 Vdc at TP-2(+) and TP-GND(-). The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.10.

4.6.3.12 Set the front panel switches as in 4.6.3.10 and apply -40.0 ± 0.005 Vdc at TP-2(+) and Tp-GND(-), The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.11.

4.6.3.13 Set the front panel switches as in 4.6.3.10 and apply -0.050 + 0.0005 Vdc at TP-2(+) and TP-GND(-). The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.12.

4.6.3.14 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 8 and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.13.

4.6.3.15 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 7 and the POWER switch to ON. Connect an external voltmeter between J3 -J(+) andJ3-X (-). The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.14.

4.6.3.16 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 6 and the POWER switch to ON. Connect an external voltmeter between J3-H (+) and J3-X (-) . The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.15.

4.6.3.17 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 5 and the PROBLEM switch to 1, 2, 4, 5 and 7 consecutively and the POWER switch to ON. Connect an external voltmeter between J3 - AA (+) and J3 - X (-). The DVM shall indicate a negative D.C. voltage for each of the designated positions of the problem switch within the limits specified in paragraph 3.4.2.16.

4.6.3.18 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 5 and the PROBLEM switch to 3 and 8 consecutively and the POWER switch to ON. Connect an external voltmeter between J3 - AA(+) and J3 - X (-). The DVM shall indicate a negative D.C. voltage for each of the designated positions of the PROBLEM switch within the limits specified in paragraph 3.2.5.17.

4.6.3.19 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 5 and the PROBLEM switch to 6 and the POWER switch to ON. Connect an external voltmeter between J3 - AA(+) and J3 - x(-). The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.18.

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4.6.3.20 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 4 and the PROBLEM switch to 1 and 3 consecutively and the POWER switch to ON. Connect an external voltmeter between J3 - x(+) and J3 - X (-). The DVM shall indicate a positive D.C. voltage for each of the designated positions of the PROBLEM switch within the limits specified in paragraph 3.5.2.19.

4.6.3.21 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 4 and the PROBLEM switch to 2 and 6 consecutively and the POWER switch to ON. Connect an external voltmeter between J3 - x (+) and J3 - X (-). The DVM shall indicate a positive D.C. voltage for each of the designated positions of the PROBLEM switch within the limits specified in paragraph 3.5.2.20.

4.6.3.22 Set DVM - CLOCK switch to SCG 1 and the SCG switch to 4 and the PROBLEM switch to 4 and 5 consecutively and the POWER switch to ON. Connect an external voltmeter between J3 - x (+) and J3 - X (-). The DVM shall indicate a positive D.C. voltage for each of the designated positions of the PROBLEM switch within the limits specified in paragraph 3.5.2.21.

4.6.3.23 Set the DVM - CLOCK switch to SCG 1 and the SCG switch to 4 and the PROBLEM switch to 7 and 8 consecutively and the POWER switch to ON. Connect an external voltmeter between J3-x(+) and J3-X(-). The DVM shall indicate a positive D.C. voltage for each of the designated positions of the PROBLEM switch within the limits specified in paragraph 3.5.2,22.

4.6.3.24 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to FV1. Apply + 4.0.0 ± 0.005 Vdc between J1-m (+) and J1 - e(-). Set the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.23.

4.6.3.25 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to 1 ø. Apply 115 ± 5% VAC, 400 ± 5% Hz between J1-c (+) and J1-d (-) with the D.C. Power Supply Ground J1-e connected to TP GND and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.24.

4.6.3.26 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to AC - SCG. Apply 115 ± 5% VAC, 400 ± 5% Hz between J3-A (+) and J3 - B (-) with the D.C. Power Supply J1-e connected to TP GND and the POWER switch set to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.25.

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4.6.3.27 Set the DVM -- CLOCK switch to VOLTAGES and the VOLTAGES switch to ø A. Apply 112 ± 5% VAC, 400 + 5% Hz between J1-V(+) and J1-U(-) with the D-C. power supply ground, J1-e, connected to TP GND and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.26.

4.6.3.28 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to ø B. Apply 115 ± 5% VAC, 400 ± 5% Hz between J1-V(+) and J1-U(-) with the D.C. power supply ground J1-e, connected to TP GND and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.27.

4.6.3.29 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to ø C. Apply 115 ± 5% VAC, 400 ± 5% Hz between J1-W(+) and J1-U(-) wit-h the D.C. power supply ground J1-e, connected to TP GND and the POWER switch to ON. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.28.

4.6.3.30 Set the RELAYS switch to K6 and the POWER switch to ON. Depress the SELF CHECK switch and the GO indicator lamp shall be illuminated as specified in paragraph 3.5.2.29.

4.6.3.31 Set the RELAYS switch to K10 and the POWER switch to ON. Depress the SELF CHECK switch and the go indicator lamp shall be illuminated as specified in paragraph 3.5.2.30.

4.6.3.32 Set the RELAYS switch to LO RATE and the POWER switch to ON. The NO-GO, indicator lamp shall extinguish if on when the RESET switch is depressed and released. The GO indicator lamp shall light and remain lit when the SELF CHECK switch is depressed and released. The GO indicator lamp shall extinguish when the RESET switch is again depressed and released as specified in paragraph 3.5.2.31.

4.6.3.33 Set the RELAYS switch to HI RATE and the POWER switch to ON. The NO-GO indicator lamp shall extinguish if on when the RESET switch is depressed and released. The GO indicator lamp shall light and remain lit when the SELF CHECK switch is depressed and released. The Go indicator lamp shall extinguish when the RESET switch is again depressed and released as specified in paragraph 3.5.2,32.

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4.6.3.34 Set the DVM - CLOCK switch to VOLTAGES and the VOLTAGES switch to FV 2. Apply the input power voltage to the test set adjusted to +27.5 ± 0.5 Vdc. Set the POWER switch to ON. The DVM shall indicate a positive D.C. voltage when the SELF CHECK switch is depressed within the limits specified in paragraph 3.5.2.33.

4.6.3.35 Set the RELAYS switch to K5 and the POWER switch to ON. Apply the normal input power voltage. Connect a multimeter , in the milliamp position, between J1-H(+) and J1-e(-). The multimeter shall indicate a milliamperage reading, when the SELF CHECK switch is depressed, within the limits specified in paragraph 3.5.2.34.

4.6.3.36 Set the same conditions as 4.6.3.35, except the multimeter is connected between J1-J(+) and J1-e(-). The multimeter shall indicate a miliamperage reading, when the SELF CHECK switch is depressed, within the limits specified in paragraph 3.5.2.35.

4.6.3.37 Set the same conditions as 4.6.3.35, except the multimeter is connected between J1-S(+) and J1-e(-) The multimeter shall indicate a miliamperage reading, when the SELF CHECK switch is depressed, within the limits specified in paragraph 3.5.2.36.

4.6.3.38 Set the same conditions as 4.6.3.35, except the multimeter is connected between J1-t (+) and J1-e(-). The multimeter shall indicate a miliamperage reading, when the SELF' CHECK switch is depressed, within the limits specified in paragraph 3.5.2.37.

4.6.3.39 Set the same conditions as 4.6.3.35, except the multimeter is connected between J1-B(+) and J1-e(-). The multimeter shall indicate a miliamperage reading, when the SELF CHECK switch is depressed, within the limits specified in paragraph 3.5.2.38.

4.6.3.40 Set the RELAYS switch to LO RATE and the DVM-CLOCK switch to LO GUN DRIVE. Apply 27.5 ± 0.5 Vdc between J1-T(+) and J1-e(-) through a NORMALLY OPEN test switch (S1) and between J1-J and J1-e(-) through a NORMALLY CLOSED test switch (S2). The DVM shall indicate a zero voltage reading within the limits specified in paragraph 3.5.2.39 when the RESET switch is depressed and released. This reading shall be adjustable with the ZERO ADJUST potentiometer. Set test switch S1 to ON momentarily and then to OFF, and set test switch S2 to OFF immediately when the NO-GO indicator lights. The DVM shall indicate a reading within the limits specified in paragraph 3.5.2.39.

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4.6.3.41 Set the RELAYS switch to LO RATE and the DVM-CLOCK switch to CLR, Apply 27.5 ± 0.5 Vdc between J1-T(+) and J1-e(-) through a NORMALLY OPEN test switch (S1) and between J1-S(+) and J1-e(-) through a NORMALLY CLOSED test switch (S2). The DVM shall indicate a D.C. voltage reading within the limits specified in paragraph 3.5.2.40, when the RESET switch is depressed and released. This reading shall be adjustable with the ZERO ADJUST potentiometer. Set test switch S1 to ON momentarily and then to OFF and set test switch S2 to OFF immediately when the NO-GO indicator lights. The DVM shall indicate a D.C. voltage within the limits specified in paragraph 3.5.2.40.

4.6.3.42 Set the RELAYS switch to LO RATE and the DVM-CLOCK switch to F.V. Apply 27.5 + 0.5 Vdc between J1-t(+) and J1-e(-) through a normally open test switch (S1) and between J1-S(+) and J1-e(-) through a normally closed test switch (S2). The DVM shall indicate a D.C. voltage reading within the limits specified in paragraph 3.5.2.41 when the RESET switch is depressed and released. This reading shall be adjustable with the ZERO ADJUST potentiometer. Set test switch S1 to ON momentarily and then to OFF, and set test switch S2 to OFF immediately when the NO-GO indicator lights. The DVM shall indicate a D.C. voltage reading within the limits specified in paragraph 3.5.2.41.

4.6.3.43 Set the RELAYS switch to LO RATE and the DVM-CLOCK switch to B.L. Apply 27.5 ± 0.5 Vdc through a NORMALLY OPEN test switch (S1) between J1-B(+) and J1-e(-) and between J1-t(+) and J1-e(-). The DVM shall indicate a D.C. voltage reading within The limits specified in paragraph 3.5.2.42, when the RESET switch is depressed and released. This reading shall. be adjustable with the ZERO ADJUST potentiometer. Set test switch S1 to ON far 2.0 + 0.3 seconds then OFF. The DVM shall indicate a D.C, voltage reading within the limits specified in paragraph 3.5.2.42.

4.6.3.44 Set the RELAYS switch to K9 and the DVM-CLOCK switch to S.T. RAMP. Connect an oscilloscope between TP-1(+) and TP-GND(-) . The DVM shall indicate a D.C. voltage reading within the limits Specified in paragraph 3.,5.2.43, when the RESET switch is depressed and released. This reading shall be adjustable with the ZERO ADJUST potentiometer. When the SELF CHECK switch is depressed and released the DVM reading and the oscilloscope reading shall indicate a voltage and pulse width within the limits specified in paragraph 3.5.2.43.

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4.6.3.45 Set the DVM - CLOCK switch to SPIN TEST. Depress and release the RESET switch. The DVM shall indicate 0.000 + 0.002 Vdc. This reading shall be adjustable with the ZERO ADJUST potentiometer. Set the SPIN TEST switch to TEST and DEPRESS the SELF CHECK switch. Allow 8 ± 0.25 seconds to elapse and then set the SPIN TEST switch to STOP. The DVM shall indicate a D.C. voltage reading within the limits specified in paragraph 3.5.2.41.

4.6.3.46 Apply 115 ± 5% VAC, 400 + 5% Hz between J1c and J1-d (-) and connect the Auxillary position indicator to Test Set connector J4 with cable W34 and set the switch inside the APU to AZIMUTH and the PROBLEM switch on the test set to 4. The APU shall indicate 192° 20′ to 192° 40′. Set the PROBLEM switch to 3. The APU shall indicate 167° 20′ to 167° 40′. Set the switch inside the APU to ELEVATION and the PROBLEM switch on the Test Set to 5. The APU shall indicate 167° 20′ to 167° 40′. Set the PROBLEM switch to 6. The APU shall indicate an angular reading within the limits specified in paragraph 3.5.2.45

4.7 Interchangeability.

4.7.1 Components. Subsequent to meeting the requirements of 3.2, 3.3, 3.4 and 3.5 in accordance with 4.3, 4.4, 4.5 and 4.6 perform this test by disassembling and reassembling each test set as follows:

a. Divide three test sets into five groups of nominated components as depicted in Table VIIA.

TABLE VIIIA. NOMINATED COMPONENTS

GROUP NO.	8438924	8438903	8438966	12004303
Test Set No. 1	a	b	c	d
Test Set No. 2	e	f	g	h
Test Set No. 3	j	k	1	m

b. Reassemble each test set and nominated component into one complete test set as depicted in Table VII B and verify that each reassembled test set will meet the performance requirements of 3.4 and 3.5 in accordance with 4.6.3.

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TABLE VIIIB

GROUP NO.	8438924	8438903	8438966	12004303
Test Set No. 1	j	f	9	m
Test Set No. 2	a	k	1	d
Test Set No. 3	e	b	c	h

4.8 Marking examination. Test sets shall be visually examined to determine compliance with 3.7.

4.9 Workmanship examination. Test sets shall be visually examined to determine compliance with 3.8.

5. Packaging.

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8438880. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. This Test Set-Air Defense Artillery AN/MWM-3 is used for the purpose of troubleshooting and verifying components of the M163 Self-Propelled and the M167 Towed, 20 Millimeter, Air Defense Artillery Gun known as the Vulcan Air Defense System (VADS). Voltages applied to this test set shall be of the Direct Current (D.C.) values specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. Requirements for First Article Submission, Including data requirements for certificates of conformance for materials.

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6.3 Inspection equipment design.

6.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

6.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.6 of this specification and with the "Test Conditions" specified in the "General requirements" of MIL-STD-810. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials and dimensions shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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